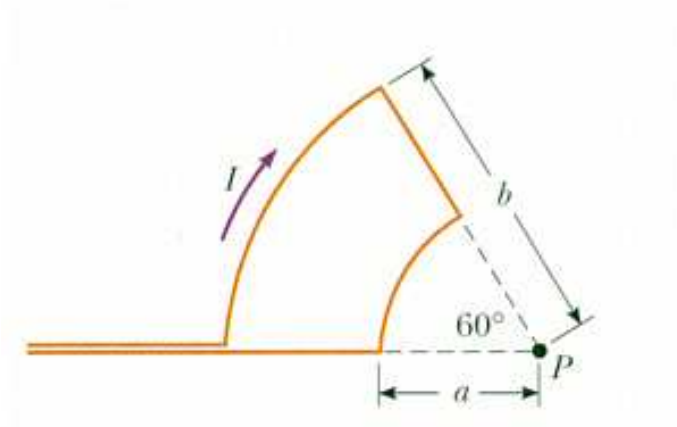


assignment 8
sb-prob3011.problem

Due date: Fri Mar 13 11:59:59 pm 2026 (EDT)

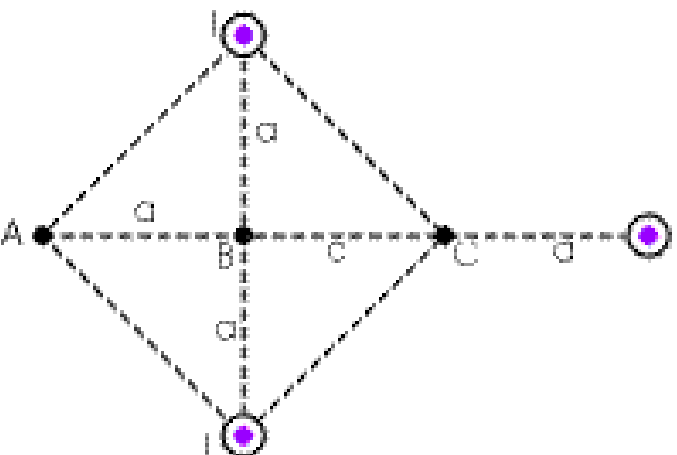
7. Consider the current-carrying loop shown in the figure, formed of radial lines and segments of circles whose center are at point P. Assume that $a=2.16\text{m}$, $b=4.32\text{m}$ and $I=3.17\text{A}$. Find the magnitude of B at P.

Tries 0/10



sb-prob3014a.problem
Three long, parallel conductors carry currents of $I = 2.08\text{A}$. The figure below is an end view of the conductors, with each current coming out of the page. If $a = 1.25\text{cm}$, determine the magnitude of the magnetic field at point A.

Tries 0/10



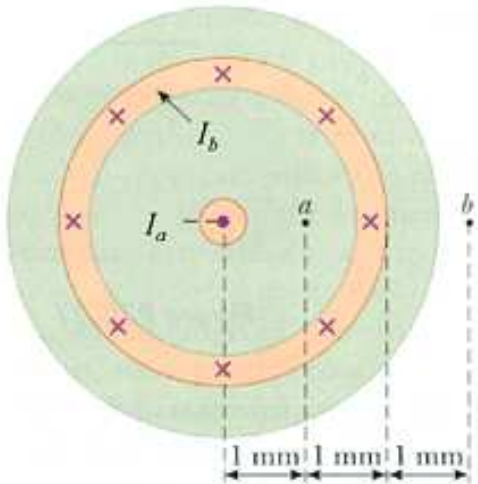
Determine the magnitude of the magnetic field at point B.

Tries 0/10

Determine the magnitude of the magnetic field at point C.

Tries 0/10

The figure shows a cross-sectional view of a coaxial cable. The center conductor is surrounded by a rubber layer, which is surrounded by an outer conductor, which is surrounded by another rubber layer. In a particular application, the current in the inner conductor is 1.22A out of the page, and the current in the outer conductor is 2.26A into the page.



Determine the magnitude of the magnetic field at point a.

Tries 0/10

Determine the magnitude of the magnetic field at point b.

Tries 0/10

sb-prob3022a.problem
The magnetic field is $1.40\mu\text{T}$ at a distance 37.0cm away from a long, straight wire. At what distance is it $0.140\mu\text{T}$?

Tries 0/10

At one instant, the two conductors in a long household extension cord carry equal 1.92A currents in opposite directions. The two wires are 2.86mm apart. Calculate the magnetic field 37.0cm way from the middle of the straight cord, in the plane of the two wires.

Tries 0/10

At what distance is it one-tenth as large?

Tries 0/10

The center wire in a coaxial cable carries current 1.92A in one direction, and the sheath around it carries current 1.92A in the opposite direction. What magnetic field does the cable create at points outside?

Tries 0/10

A cylindrical conductor of radius $R = 2.64\text{cm}$ carries a current of $I = 2.68\text{A}$ along its length; this current is uniformly distributed throughout the cross-section of the conductor. Calculate the magnetic field midway along the radius of the wire (that is, at $r = R/2$).

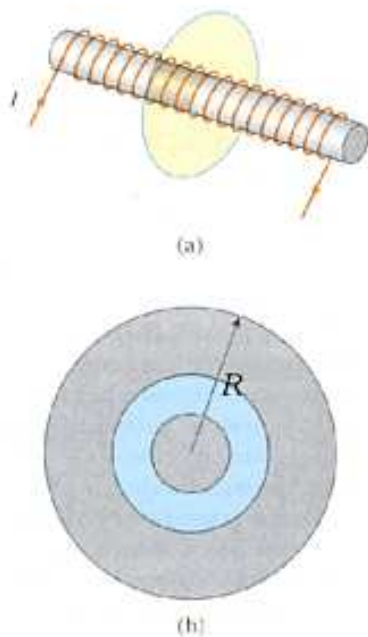
Tries 0/10

Calculate the distance beyond the surface of the conductor at which the magnitude of the magnetic field has the same value as the magnitude of the field at $r = R/2$.

Tries 0/10

sb-prob3034a.problem

A solenoid $2.86\text{E-}2\text{m}$ in diameter and 0.381m long has 346 turns and carries 12.0A .



Calculate the flux through the surface of a disk of radius $5.00\text{E-}2\text{m}$ that is positioned perpendicular to and centred on the axis of the solenoid.

Tries 0/10

Figure b) shows an enlarged end view of the same solenoid as in the last question. Calculate the flux through the blue area, which is defined by an annulus that has an inner radius of 0.311cm and an outer radius of 0.622cm .

Tries 0/10

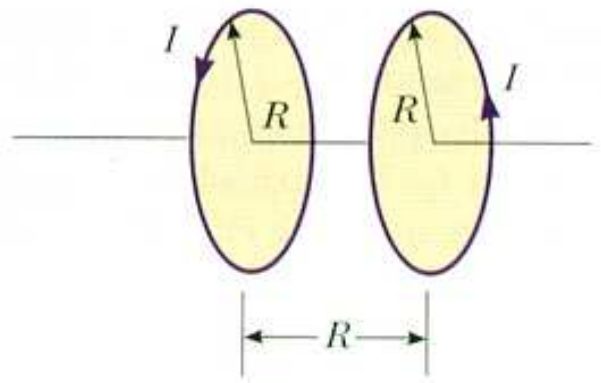
sb-prob3042.problem

A uniform ring with a radius of 2.06cm and a total charge of 6.20uC rotates with a constant angular speed of 3.93rad/s around an axis perpendicular to the plane of the ring and passing through its center. What is the magnetic moment of the rotating ring?

sb-prob3058.problem

Two identical, flat, circular coils of wire each have 102 turns and a radius of 0.506m . The coils are arranged as a set of Helmholtz coils, parallel and with a separation of 0.506m . If each coil carries a current of 10.0A , determine the magnitude of the magnetic field at a point on the common axis of the coils and halfway between them.

Tries 0/10



sb-prob3066.problem

8. An infinitely long, straight wire carrying a current $I_1 = 10.5\text{A}$ is partially surrounded by a loop. The loop has a length $L = 4.02\text{m}$ and a radius $R = 0.272\text{m}$ and carries a current $I_2 = 13.1\text{A}$. The axis of the loop coincides with the wire. Calculate the magnitude of the force exerted on the loop.

Tries 0/10

